

Work Breakdown Structure

Vietnamese Restaurant Chain
Mobile Ordering Application

Project	Vietnamese Restaurant Mobile Ordering App
Client	Vietnamese Restaurant Chain (5 Locations)
Date	March 10, 2026
Version	1.0
Status	Draft

PHASES	WORK PACKAGES	TASKS	PLATFORMS
9	27	135	iOS + Android

Project Overview

This WBS defines the full scope of work for building a cross-platform mobile ordering application for a Vietnamese restaurant chain with 5 locations. The system encompasses a customer-facing app (iOS and Android), a staff-facing tablet dashboard for real-time order management, backend services, an admin portal, and supporting infrastructure.

Key capabilities include menu browsing by location, online payment processing, real-time order tracking, a loyalty points program, and multi-location operations management. The project follows a phased rollout strategy starting with a single pilot location before expanding to all 5 sites.

Key Assumptions

- Cross-platform development using React Native or Flutter (single codebase for iOS and Android).
- Pickup and dine-in ordering only; delivery is deferred to a future phase.
- Payment processing via Stripe or equivalent PCI-compliant gateway.
- Cloud-hosted infrastructure with auto-scaling for peak meal times.
- Staff dashboard optimized for tablet form factors (iPad and Android tablets).
- Phased rollout: pilot at 1 location, then expand to remaining 4.

WBS Detail Breakdown

1.0 Project Management & Planning	
1.1	Project Initiation
	1.1.1 – Define project charter and objectives
	1.1.2 – Stakeholder identification and analysis
	1.1.3 – Establish project governance structure
	1.1.4 – Define success criteria and KPIs
1.2	Planning & Estimation
	1.2.1 – Create detailed project schedule (Gantt)
	1.2.2 – Resource planning and team allocation
	1.2.3 – Budget estimation and approval
	1.2.4 – Risk assessment and mitigation plan
	1.2.5 – Communication plan (stakeholders, team)
1.3	Vendor & Tool Selection
	1.3.1 – Evaluate cross-platform frameworks (React Native / Flutter)
	1.3.2 – Select payment gateway provider
	1.3.3 – Select cloud infrastructure (AWS / GCP / Azure)
	1.3.4 – Select CI/CD and DevOps tooling

2.0 Requirements & Design

2.1 Requirements Gathering

- 2.1.1 – Customer journey mapping (order, pay, pickup)
- 2.1.2 – Staff workflow analysis (order receipt, prep, completion)
- 2.1.3 – Menu data structure requirements (5 locations, variations)
- 2.1.4 – Loyalty program rules definition
- 2.1.5 – Multi-location business rules (hours, pricing, availability)
- 2.1.6 – Compliance requirements (PCI-DSS, data privacy)

2.2 UX/UI Design

- 2.2.1 – Information architecture and navigation flow
- 2.2.2 – Wireframes: Customer app (all screens)
- 2.2.3 – Wireframes: Staff tablet dashboard
- 2.2.4 – Visual design system (brand colors, typography, icons)
- 2.2.5 – High-fidelity mockups: Customer app
- 2.2.6 – High-fidelity mockups: Staff dashboard
- 2.2.7 – Interactive prototype and usability testing
- 2.2.8 – Design review and sign-off

2.3 Technical Architecture

- 2.3.1 – System architecture design (client, API, DB, services)
- 2.3.2 – API contract definition (OpenAPI / GraphQL schema)
- 2.3.3 – Database schema design
- 2.3.4 – Real-time communication architecture (WebSocket / SSE)
- 2.3.5 – Security architecture (auth, encryption, tokenization)
- 2.3.6 – Architecture review and approval

3.0 Customer Mobile App (iOS & Android)

3.1 Authentication & Profile

- 3.1.1 – User registration (email, phone, social login)
- 3.1.2 – Login / logout / session management
- 3.1.3 – Profile management (name, phone, addresses)
- 3.1.4 – Password reset and account recovery
- 3.1.5 – Push notification opt-in and preferences

3.2 Restaurant & Menu Browsing

- 3.2.1 – Location selector (map view, list view, GPS auto-detect)
- 3.2.2 – Menu display with categories and search
- 3.2.3 – Item detail view (photos, description, allergens, spice level)
- 3.2.4 – Customization options (add-ons, modifications, notes)
- 3.2.5 – Menu availability by location and time of day

3.3 Cart & Ordering

- 3.3.1 – Add/remove/edit items in cart
- 3.3.2 – Cart summary with subtotal, tax, fees
- 3.3.3 – Order type selection (pickup, dine-in)
- 3.3.4 – Scheduled order (pick future time slot)
- 3.3.5 – Promo code / coupon application
- 3.3.6 – Order submission and confirmation

3.4 Payment Integration

- 3.4.1 – Credit/debit card payment (Stripe / Braintree)
- 3.4.2 – Apple Pay integration
- 3.4.3 – Google Pay integration
- 3.4.4 – Save payment methods for future use
- 3.4.5 – Payment receipt and email confirmation
- 3.4.6 – Refund handling (customer-side display)

3.5 Order Tracking

- 3.5.1 – Real-time order status updates (placed, preparing, ready)
- 3.5.2 – Push notifications for status changes
- 3.5.3 – Estimated prep time display
- 3.5.4 – Order history with reorder capability

3.6 Loyalty Program

- 3.6.1 – Points balance and tier display

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| 3.6.2 – Points earning on orders |
| 3.6.3 – Rewards catalog and redemption |
| 3.6.4 – Points history and transaction log |
| 3.6.5 – Referral program (invite friends, earn bonus) |

4.0 Staff Tablet Dashboard

4.1 Staff Authentication & Setup

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| 4.1.1 – Staff login (PIN or credentials) |
| 4.1.2 – Role-based access (manager, kitchen, counter) |
| 4.1.3 – Location binding and shift configuration |

4.2 Real-Time Order Management

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| 4.2.1 – Live order queue (new orders appear in real-time) |
| 4.2.2 – Order detail view with items, mods, notes |
| 4.2.3 – Order status update (accept, preparing, ready, completed) |
| 4.2.4 – Order prioritization and time tracking |
| 4.2.5 – Audio/visual alerts for new orders |
| 4.2.6 – Bulk order actions (mark multiple ready) |

4.3 Menu & Operations Management

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| 4.3.1 – Toggle item availability (86'd items) |
| 4.3.2 – Adjust prep time estimates |
| 4.3.3 – Pause/resume online ordering |
| 4.3.4 – Daily order summary and reports |

5.0 Backend Services & Infrastructure

5.1 Core API Development

- 5.1.1 – User service (registration, auth, profile)
- 5.1.2 – Menu service (CRUD, categories, availability)
- 5.1.3 – Order service (create, update, status, history)
- 5.1.4 – Payment service (process, refund, webhook handling)
- 5.1.5 – Loyalty service (points calculation, rewards, tiers)
- 5.1.6 – Notification service (push, email, SMS)
- 5.1.7 – Location service (store info, hours, geo-queries)

5.2 Real-Time Infrastructure

- 5.2.1 – WebSocket server for order status streaming
- 5.2.2 – Event-driven architecture (order lifecycle events)
- 5.2.3 – Message queue setup (RabbitMQ / SQS) for async processing

5.3 Database & Storage

- 5.3.1 – Primary database setup (PostgreSQL)
- 5.3.2 – Caching layer (Redis) for menus, sessions
- 5.3.3 – Image/asset storage (S3 / CDN)
- 5.3.4 – Database migration and seed scripts

5.4 Cloud Infrastructure & DevOps

- 5.4.1 – Infrastructure as Code (Terraform / CloudFormation)
- 5.4.2 – CI/CD pipeline (build, test, deploy automation)
- 5.4.3 – Environment setup (dev, staging, production)
- 5.4.4 – Auto-scaling and load balancing configuration
- 5.4.5 – Logging, monitoring, alerting (Datadog / CloudWatch)
- 5.4.6 – Backup and disaster recovery plan

6.0 Admin Portal (Web)

6.1 Menu Management

- 6.1.1 – CRUD for menu items, categories, modifiers
- 6.1.2 – Photo upload and management
- 6.1.3 – Pricing management (per-location overrides)
- 6.1.4 – Bulk import/export (CSV/Excel)

6.2 Business Operations

- 6.2.1 – Multi-location dashboard (orders, revenue, trends)
- 6.2.2 – Staff account management
- 6.2.3 – Loyalty program configuration (rules, tiers, rewards)
- 6.2.4 – Promo code and campaign management
- 6.2.5 – Customer data and analytics
- 6.2.6 – Revenue reports and export

7.0 Quality Assurance & Testing

7.1 Testing Strategy & Automation

- 7.1.1 – Test plan creation (scope, approach, environments)
- 7.1.2 – Unit test coverage (backend services, frontend components)
- 7.1.3 – Integration testing (API, payment, notification flows)
- 7.1.4 – End-to-end testing (customer ordering flow, staff flow)
- 7.1.5 – Performance and load testing (concurrent orders, peak hours)

7.2 Specialized Testing

- 7.2.1 – Security testing and penetration testing
- 7.2.2 – Payment flow testing (PCI compliance verification)
- 7.2.3 – Multi-device and OS version testing
- 7.2.4 – Accessibility testing (WCAG compliance)
- 7.2.5 – Offline/poor connectivity behavior testing
- 7.2.6 – User acceptance testing (UAT) with restaurant staff

8.0 Deployment & Launch

8.1 App Store Submission

- 8.1.1 – Apple App Store listing (screenshots, description, metadata)
- 8.1.2 – Google Play Store listing
- 8.1.3 – App review and approval process
- 8.1.4 – Beta testing program (TestFlight / Play Console)

8.2 Rollout Strategy

- 8.2.1 – Pilot launch at 1 location
- 8.2.2 – Staff training and onboarding materials
- 8.2.3 – Feedback collection and iteration
- 8.2.4 – Phased rollout to remaining 4 locations
- 8.2.5 – Go-live checklist and cutover plan

9.0 Post-Launch & Operations

9.1 Support & Maintenance

- 9.1.1 – Bug triage and hotfix process
- 9.1.2 – Customer support integration (in-app chat / help center)
- 9.1.3 – Performance monitoring and optimization
- 9.1.4 – Regular dependency and security updates

9.2 Growth & Iteration

- 9.2.1 – Analytics review and feature prioritization
- 9.2.2 – A/B testing framework for UX improvements
- 9.2.3 – App update release cadence (bi-weekly / monthly)
- 9.2.4 – Roadmap planning for V2 features (delivery, catering, etc.)

Phase Summary

WBS ID	Phase	Work Packages	Tasks
1.0	Project Management & Planning	3	13
2.0	Requirements & Design	3	20
3.0	Customer Mobile App (iOS & Android)	6	31
4.0	Staff Tablet Dashboard	3	13
5.0	Backend Services & Infrastructure	4	20
6.0	Admin Portal (Web)	2	10
7.0	Quality Assurance & Testing	2	11
8.0	Deployment & Launch	2	9
9.0	Post-Launch & Operations	2	8
	TOTAL	27	135

Notes

1. This WBS covers scope definition only. Effort estimates, resource assignments, and schedule dependencies are maintained in the project plan.
2. Work packages may be further decomposed during sprint planning as the team refines acceptance criteria for each item.
3. The WBS follows a product-oriented decomposition. Cross-cutting concerns (security, performance, accessibility) are embedded within each phase rather than isolated.
4. Delivery integration (e.g., DoorDash, UberEats) and catering features are explicitly excluded from V1 scope and reserved for the V2 roadmap.